

Mechanism of Heterogeneous Bubble Nucleation in Polymer Blend Foaming

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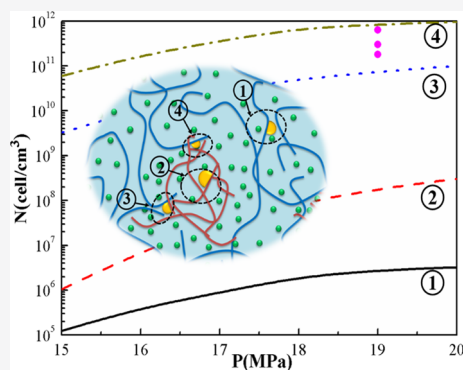
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ABSTRACT: A three-dimensional heterogeneous bubble nucleation model is constructed to provide a reasonable explanation at the molecular level for the foaming mechanism of polypropylene (PP) and polystyrene (PS) blends. CO₂ solubilities and supersaturation ratios are quantitatively calculated to help interpret the contribution of each phase of the blend in the CO₂ dissolution stage. The spatial density profiles of polymer/CO₂ binary melt around different polymer chains are presented to give an intuitive perspective to the thermodynamic driving force. The predicted interfacial tension and contact angles of critical bubbles provide valid evidence to distinguish the wettability of CO₂ in different regions. The values of predicted free-energy barriers, critical radii, and nucleation number densities imply that bubbles that nucleate in the PP and PS blend interfacial region attached to the PS-rich phase achieve the smallest size and largest number density. The reliability of the theoretical model has been tested by partial available experimental data.



1. INTRODUCTION

Thermoplastic foams with excellent microcellular networks show good application prospects because of their many superior characteristics.^{1–8} However, some inherent drawbacks, such as low melt strength and heat-resistance temperature, slow crystallization rate, and poor toughness of single polymers, restricted the preparation and application of polymer foams.^{9–14} Polymer blending is a simple, effective, and economical means for obtaining materials with desired properties,^{15–20} which not only makes up for the deficiency of single-polymer systems but also remarkably enhances the foaming ability.

The reported experiments reveal that polymer blends show different foaming behaviors compared to single polymers, where the CO₂ solubility, CO₂ diffusion coefficients, cell density, cell size, and pore size distribution are different in each phase of the blend.^{15,21–27} The different geometric and chemical properties of the blend components lead to their different roles in the foaming process, which add to the complexity of cell structure prediction. Although remarkable progress has been achieved in enhancing cell nucleation and ensuring cell growth to prepare high porosity polymer blend foams, the cell size, cell density, and pore size distribution of polymer foams still limit the properties in some special applications. This is mainly attributed to the experimental research studies that just analyze the polymer blend foaming process at the macrolevel, which cannot quantitatively calculate the contribution of each polymer of the blend to CO₂ dissolution and bubble nucleation processes. Hence, studies on polymer blend foaming behavior are imperative for preparing desired cell structures to meet the end uses.

Polymer blend is used as a means to control the foam cell morphology for a long time, but the mechanism behind it has been rarely systematically studied. Theoretical rules are effective means for underlining the nucleation mechanism that is responsible for the final structure and properties of polymer foams. Various hypotheses have been put forward to explain the bubble nucleation mechanism in polymer blend foaming systems. Although these hypotheses are invaluable in providing accumulated experiences, they need further studies to verify the reliability. On the other hand, density functional theory (DFT) has achieved good performance in polymer foaming systems.^{28–31} Within the theoretical framework, a bubble nucleation occurs in the actual environment, involving the local supersaturation of dissolved CO₂ and subsaturation of the polymer matrix as well as the interfacial tensions of bubble–melt–interface three-phase contact. The influence of experiment conditions, component characteristics, and interfacial structures on bubble nucleation can be calculated by minimizing the grand potential of the system. This shows a good application prospect for the prediction of blend foaming systems.

In this work, we try to provide a comprehensive mechanism explanation for polymer blend foaming systems, where PP and

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PS blend systems are taken as examples. Within the framework of three-dimensional density functional theory (3D-DFT), the free-energy functional is derived from the interactions of polymer/CO₂ binary melt. The CO₂ solubility and supersaturation degree in PP and PS are quantitatively calculated to help interpret the contribution of each phase of the blend in the CO₂ dissolution stage. To further give an insight into the thermodynamic driving forces for bubble nucleation in polymer blend foaming systems, we analyze the interfacial density distribution and interfacial tension of the polymer/CO₂ binary melt around the polymer matrix in different domains. The microstructures of critical bubbles nucleated from different domains are then presented to provide a direct view of bubble morphologies. Finally, the critical radii, free-energy barriers, and bubble number density are calculated, which decipher the foaming process by effective data. This work is expected to provide a theoretical basis for understanding the heterogeneous nucleation mechanism of polymer blend foams.

2. THEORETICAL SECTION

The schematic drawing for the PP/PS/CO₂ blend system is shown in Figure 1. There are mainly four types of domains in

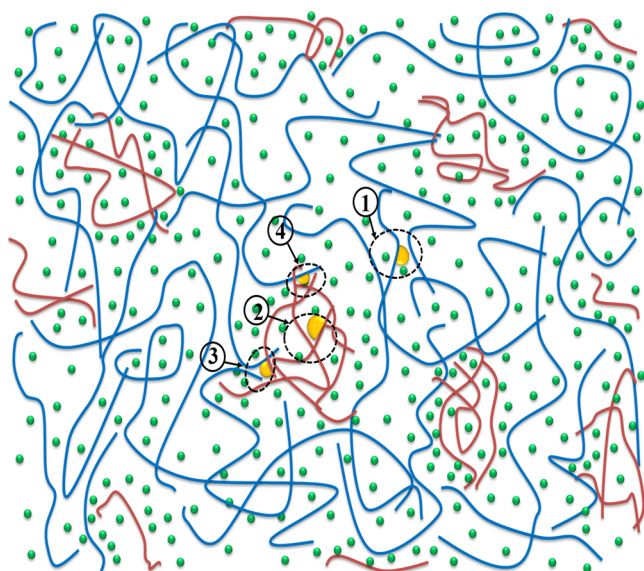


Figure 1. Schematic drawing for the PP/PS/CO₂ blend system. The blue lines represent PP chains, the red lines are PS chains, the green balls are dissolved CO₂ molecules, and the yellow balls stand for bubbles that may be nucleated from the referred domains in the foaming process.

the PP/PS/CO₂ blend system: (1) the PP domain referred to as PP-PP/CO₂ (marked by ① in Figure 1), where PP chains provide heterogeneous nucleation sites and PP and CO₂ constitute the binary melt to influence the thermodynamic properties of dissolved CO₂ molecules; (2) the PS domain referred to as PS-PS/CO₂ (marked by ② in Figure 1), where PS chains provide heterogeneous nucleation sites and PS and CO₂ constitute the binary melt to influence the thermodynamic properties of dissolved CO₂ molecules; (3) the PP and PS blend interfacial region attached to the PP-rich phase referred to as PS-PP/CO₂ (marked by ③ in Figure 1), where PS chains provide heterogeneous nucleation sites and PP and CO₂ constitute the binary melt to influence the thermodynamic

properties of dissolved CO₂ molecules; and (4) the PP and PS blend interfacial region attached to the PS-rich phase referred to as PP-PS/CO₂ (marked by ④ in Figure 1), where PP chains provide heterogeneous nucleation sites and PS and CO₂ constitute the binary melt to influence the thermodynamic properties of dissolved CO₂ molecules. Different domains in the blend system provide different opportunities for bubble nucleation, and the yellow balls stand for bubbles that may be nucleated in the referred domains during the foaming process.

In the theoretical model, the grand potential, $\Omega[\{\rho_\alpha(\mathbf{r})\}]$, can be generally expressed as

$$\begin{aligned} \Omega[\{\rho_\alpha(\mathbf{r})\}] = & k_B T \sum_\alpha \int d\mathbf{r} \rho_\alpha(\mathbf{r}) [\ln(\rho_\alpha(\mathbf{r})) - 1] \\ & + F^{\text{ex}}[\{\rho_\alpha(\mathbf{r})\}] + k_B T \\ & \sum_\alpha \int d\mathbf{r} \rho_\alpha(\mathbf{r}) (U_\alpha(\mathbf{r}) - \mu_\alpha) \end{aligned} \quad (1)$$

in which the first and second terms on the right-hand side represent the ideal and excess free-energy contributions, respectively. α represents a CO₂ molecule or a site in a polymer chain, $U_\alpha(\mathbf{r})$ is the external potential of a nucleation site, and μ_α stands for the bulk chemical potential of component α . The excess free energy can be generalized as

$$\begin{aligned} F^{\text{ex}}[\rho_1(\mathbf{r}), \rho_2(\mathbf{r})] = & F^{\text{hs}}[\rho_1(\mathbf{r}), \rho_2(\mathbf{r})] + F^{\text{att}}[\rho_1(\mathbf{r}), \rho_2(\mathbf{r})] \\ & + F^{\text{chain}}[\rho_1(\mathbf{r}), \rho_2(\mathbf{r})] \\ & + F^{\text{conf}}[\rho_1(\mathbf{r}), \rho_2(\mathbf{r})] \end{aligned} \quad (2)$$

to account for the contributions arising from hard-sphere repulsion, dispersive attraction, chain connectivity, and chain conformation, respectively. Detailed information is given in the Supporting Information (SI). $U_\alpha(\mathbf{r})$ is generated by the polymer matrix, which works as a heterogeneous nucleation site. It can be calculated by

$$U_\alpha(\mathbf{r}) = \sum_p U_{p\alpha}(\mathbf{r}) \quad (3)$$

where $U_{p\alpha}(\mathbf{r})$ contains the potentials of polymer–CO₂ and polymer–polymer interactions. For heterogeneous bubble nucleation research, the modeling PP and PS surface needs to be constructed to mimic the real systems. A molecular dynamic (MD) simulation method is applied to construct the amorphous PP and PS surface. CO₂ is modeled as a sphere molecule, and PP and PS chains consist of 200 repeat units. The information of PP and PS for simulation is provided in the SI, and the constructed processes are also available. The reliability of the force field parameters^{32–34} for CO₂ and polymers has already been verified elsewhere.^{28–30}

By minimizing the grand potential, the density profiles of a bubble or interface can be determined with

$$\rho_\alpha(\mathbf{r}) = \exp \left(\mu_\alpha - \frac{\delta(F^{\text{ex}}[\{\rho_\alpha(\mathbf{r})\}])}{\delta \rho_\alpha(\mathbf{r})} - \beta U_\alpha(\mathbf{r}) \right) \quad (4)$$

For the bubble displaying 3D characteristics, we adopt the Cartesian coordinate with the initial density profile

$$\rho_{\alpha}(x, y, z) = \begin{cases} 0 & z \leq z_0 \\ \rho_{v,\alpha} & r \leq R_0 \\ \rho_{s,\alpha} & r > R_0 \end{cases} \quad (5)$$

where z_0 is the position of the polymer matrix for the boundary effect, R_0 is the initial radius of the nucleus with r being the distance to the center of the spherical cap, $\rho_{v,\alpha}$ is the density of component α at 0.1 MPa, and $\rho_{s,\alpha}$ is the density of component α around the bubble at a supersaturation (or subsaturation) state. In a 3D space, the density distributions of CO_2 and polymer melt around the polymer matrix are calculated on a grid of $256 \times 256 \times 256$ points, which gives a grid resolution of $0.1\sigma_{\text{CO}_2}$ on each axis. The size is chosen so that the density distributions are not affected by edge effects. The traditional Picard iteration is applied to obtain the density profiles, and the iteration procedure is repeated until the average fractional difference between the old and new densities at any 3D grid point is less than 10^{-4} . The position of a bubble interface is then determined by the subsaturation density of a polymer with $\rho_{s,\text{polymer}}/2$. A “pseudo” grand canonical ensemble method³⁵ is applied to calculate the final density profiles and subsequent thermodynamic properties of the critical nuclei.

The interfacial tension γ is the excess free energy for polymers and CO_2 binary melt around a polymer matrix, which includes the contributions in terms of CO_2 – CO_2 , CO_2 –polymer, and polymer–polymer interactions. It is written as

$$\gamma = \Delta\Omega/A = [\Omega[\rho(\mathbf{r})] - \Omega(\rho_{\text{bulk}})]/A \quad (6)$$

where $\Omega(\rho_{\text{bulk}})$ is the grand potential at the equilibrium state and A is the interfacial area.

Finally, the stationary nucleation rate J [$\text{cm}^{-3} \text{s}^{-1}$] for bubble nucleation can be expressed as

$$J = J_0 \exp(-\Delta\Omega/k_B T) \quad (7)$$

where the pre-exponential factor J_0 contains the information of bubble–melt–interface ternary phase contact (details are listed in the SI). The reliability of J_0 has been verified in our previous work.^{28,29} For simplicity, the bubble nucleation process is considered in 1 s. Accordingly, the total number of nuclei N [$\text{nuclei}/\text{cm}^3$] can be determined by integrating the nucleation rate over the time period.

3. RESULTS AND DISCUSSION

The studied temperature is selected at 452.15 K to reveal the foaming mechanism of the PP/PS blend system, where there is available experimental data of nucleation number densities to verify the reliability of the used theoretical model. Taking the foaming that occurs at ambient pressure (0.1 MPa) as an example, we analyze the bubble nucleation process in detail. After introducing supercritical CO_2 into the PP/PS blend system, the CO_2 phase and polymer– CO_2 mixed phase in different domains will reach their own equilibrium. The solubility can be predicted based on the fact that the corresponding component in the two phases has the same chemical potential at equilibrium. As shown in Figure 2, the deviation between predicted values and experimental data^{36,37} at 373.15 K for PP is 4.73% and that for PS is 5.28%. This implies the reliability of the theoretical model. Comparing the calculated values, we can find that CO_2 solubility in PP is higher than that in PS under the same condition. The presence

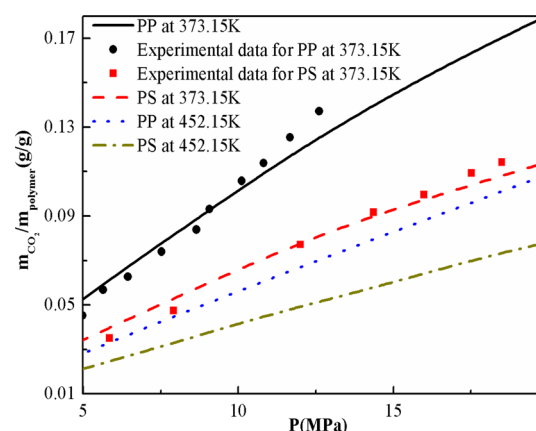


Figure 2. CO_2 solubilities in PP and PS at different temperatures. The average molecular masses of PP and PS in the experimental references^{36,37} are 4.20×10^4 and 7.03×10^4 g/mol, respectively.

of a benzene ring in the PS system takes primary responsibility. The big occupied volume of the benzene ring compresses the available free volume in the PS matrix, which is not conducive to the dissolution of CO_2 molecules in the PS matrix. On the other hand, the dissolution of CO_2 in the polymer matrix is related to the fluidity of the polymer chain. When the solubility of CO_2 reaches a certain value, the fluidity of the polymer chain plays an important role in adsorption. Since the PP chain has better fluidity than the PS chain, the adsorption capacity of CO_2 in the PP system is enhanced again. Therefore, more CO_2 molecules accumulate in the PP domain at high pressure, which will provide a reserve for subsequent bubble nucleation.

After CO_2 molecules have dissolved sufficiently, the PP/PS/ CO_2 system will perform an expansion process, keeping the mass fraction of each component constant in the decompression stage. Figure 3 shows the supersaturation ratios for

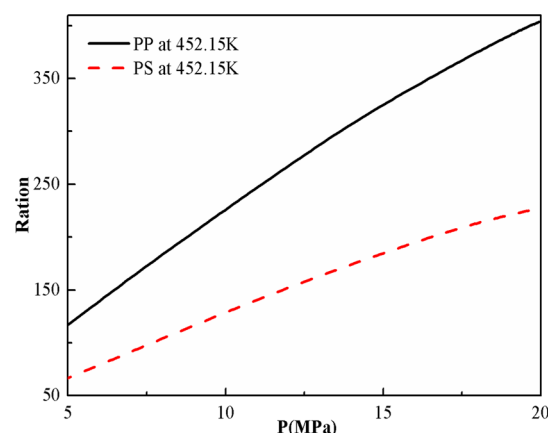


Figure 3. Supersaturation ratios of CO_2 in the PP and PS matrix, which are the ratios of supersaturated CO_2 density to its density at 0.1 MPa.

different domains in the blend system after the pressure declines from the initial state to 0.1 MPa. It is obvious that the values increase with pressure, the difference between PP and PS domains becomes obvious as pressure increases, and the supersaturation ratios of the PP domain reach twice those of the PS domain at 20 MPa. The supersaturation ratios are closely related to the solubility of CO_2 and the properties of polymer melt. Higher CO_2 solubility and better flexibility of PP

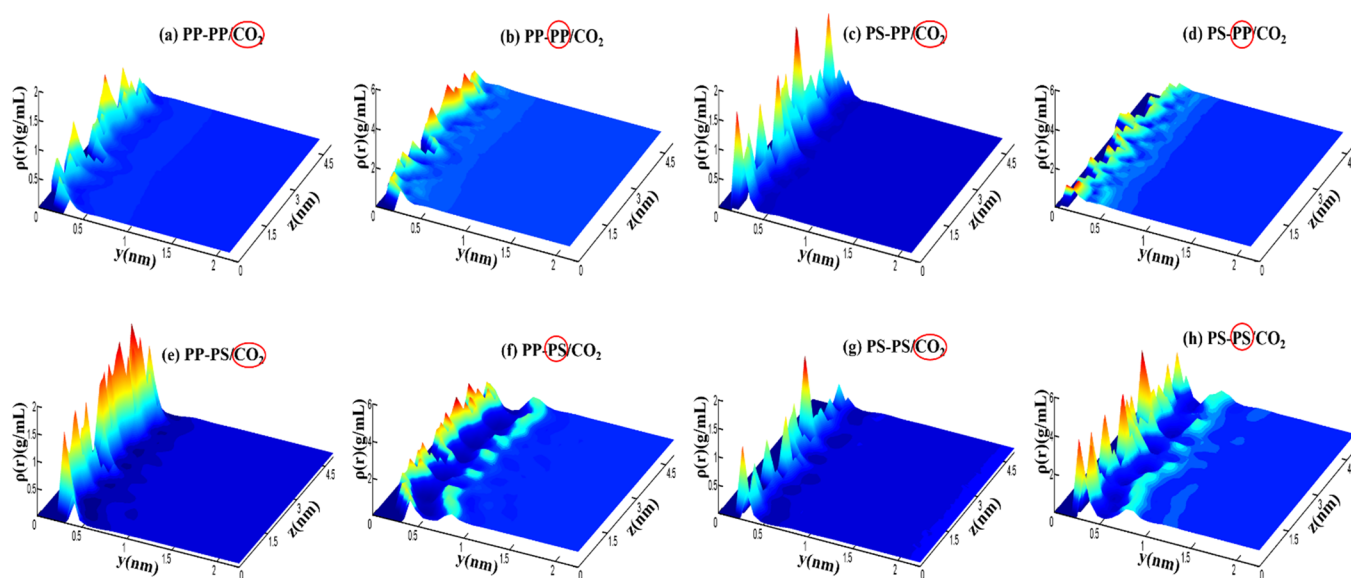


Figure 4. Two-dimensional cuts to simultaneously display the spatial density distributions of the melted polymer and CO_2 around polymer chains for the referred four domains. The red circles in the subfigures point out the shown component. The mass fractions of CO_2 and polymers are determined at 19.0 MPa.

chains account for the higher supersaturation ratios in PP domains. Higher supersaturation ratio means more CO_2 molecules available, where the PP domains provide the possibility for the formation of small bubbles and high cell density in the blend system.

Interfacial microstructure analysis is an effective tool to interpret the surface wetting mechanism for different interfaces. Figure 4 displays the spatial density profiles of CO_2 and polymers around PP or PS chains for the domain of PP-PP/ CO_2 , PS-PS/ CO_2 , PS-PP/ CO_2 , and PP-PS/ CO_2 . On the whole, one can immediately see the obvious high density peaks of CO_2 around polymer chains, and the density distributions of PP and PS close to the interface show multilayer adsorptions, where the density distributions vary from peak to trough, implying the transformation of local adsorption/desorption. As shown in Figure 4a,b, the PP and CO_2 display lower adsorption around PP chains in the PP domain. When PP melts is in contact with PP chains, it follows the same compatibility principle, and this is not conducive to the adsorption of CO_2 at the interface. However, the comparison of Figure 4a,g as well as Figure 4b,h implies that more CO_2 molecules accumulate around PP chains in the PP domain, and the PS density around PS chains in the PS domain is stronger than the PP density around PP chains in the PP domain. This is mainly attributed to the benzene ring in the PS chain that leads to small free volume and poor flexibility, where the better flexibility of the PP chain is more conducive to the interfacial accumulation of CO_2 molecules. The adsorption peaks in Figure 4c,e are much stronger than those in Figure 4a,g. This is mainly attributed to the depletion effect induced by the interaction of PP and PS, which enhances the osmotic pressure, driving more CO_2 molecules to occupy the interfacial space. One can see that much more voids are displayed in Figure 4f than that in Figure 4d, resulting in a multitude of CO_2 molecules squeezed into the blend interface region in Figure 4e. These phenomena are induced by the synergistic effects of chemical characteristics and topological structures of PP and PS, where the PS melts resist from the PP surface, facilitating the accumulation of CO_2 molecules.

Summarizing the local density distribution, one can find that there exists a larger CO_2 concentration gradient across the PP/PS interface at high pressure, and the CO_2 concentration gradient causes the diffusion of CO_2 from the PP phase to and across the PP/PS interface and finally to the PS phase. The much higher CO_2 solubility in the PP domain referred in Figure 2 provides the possibility for the migration of CO_2 molecules. The predicted results are consistent with those explained in the experimental research.³⁸ Therefore, CO_2 local density near the blend interface increases dozens of times, while the local density of polymer melt decreases dozens of times, resulting in further increments of CO_2 supersaturation and polymer subsaturation in the interfacial region. The microscopic interface distribution subfigures of different domains provide an intuitive mechanism explanation for the distribution of CO_2 in the blend system after depressurization.

The above density fluctuations correlate to interfacial energy or interfacial tension, which can be calculated by eq 6. Figure 5

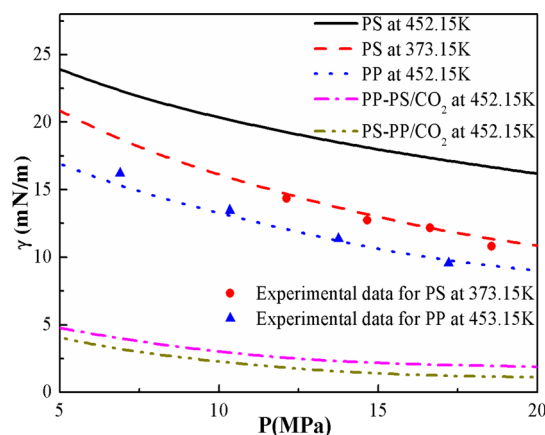


Figure 5. Interfacial tensions of polymer/ CO_2 binary melts in contact with the surfaces of polymer chains. The average molecular masses of PP and PS in the experimental references^{39,40} are 1.05×10^5 and 1.71×10^5 g/mol, respectively.

presents the calculated interfacial tension for the mentioned four situations in Figure 4. For the limitation of available experimental data, the predicted values for PP-PP/CO₂ at 452.15 K are examined by the experimental data³⁹ at 453.15 K, and the deviation is 3.49%. The deviation between calculated values and experimental data⁴⁰ for PS-PS/CO₂ at 373.15 K is 2.535%. The calculated value generally agrees with the experimental ones, which indicates that the theory provides quantitatively reasonable description for the interfacial microstructures and properties of the PP/PS/CO₂ blend system. One sees that the values decrease as the pressure increases, the interfacial tension for PP-PP/CO₂ is lower than that for PS-PS/CO₂, the interfacial tension for PS-PP/CO₂ and PP-PS/CO₂ achieves much lower values, and values for PS-PP/CO₂ are the smallest. The high interfacial tension of PS itself determines the value of interfacial tension associated with it. Combining with Figure 4, we can find that the accumulated CO₂ molecules greatly reduce the interfacial tension of PS-PP/CO₂ and PP-PS/CO₂, where higher CO₂ density corresponds to lower interfacial tension, implying better wettability. These predictions offer a helpful guideline for analyzing bubble nucleation in different domains.

The formations of new bubbles nucleated from the mixture of CO₂ and polymers in different domains are then investigated, which is crucial for interpreting the foaming mechanism of the PP/PS blend system. If the CO₂ molecules are adequate, bubbles will nucleate in different domains under their own metastable equilibrium after the decompression process. Figure 6 shows the constrained free-energy curves of

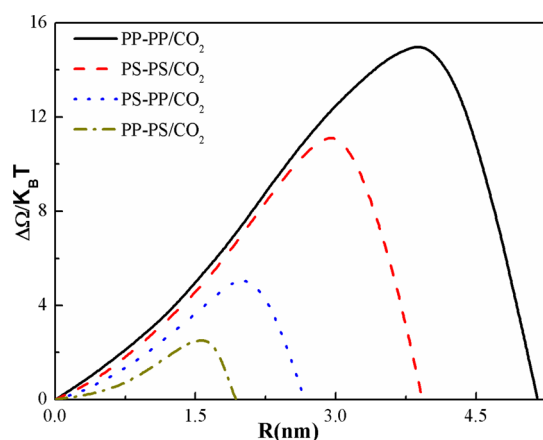


Figure 6. Constrained free-energy curves of growing bubbles nucleated in different domains. The mass fractions of CO₂ and polymers are determined at 19.0 MPa.

the growing bubbles in different domains, and the mass fractions of CO₂ and polymers are also determined at 19.0 MPa. As shown, any initial radius corresponds to its free energy, and the critical nucleation radius and nucleation barrier are obtained at the highest point. The critical nucleation barriers for PP-PP/CO₂, PS-PS/CO₂, PS-PP/CO₂, and PP-PS/CO₂ are 15.02, 11.17, 5.11, and 2.53, respectively. The nucleation radii are 3.89, 2.96, 2.00, and 1.57 nm, respectively. Therefore, the addition of PS in the PP/CO₂ system leads to reductions in energy consumption of 25.63% in the PS domain and 65.98–83.16% in the blend interfacial domain, and the shrinkage values of critical radius are 23.91 and 48.59–59.64%, respectively. The results suggest that bubbles nucleate more easily in the blend interfacial domain than in the PP or PS

domain, where the blend interface attached to the PS-rich phase provides a more convenient environment for bubble nucleation.

In order to give an intuitive understanding for the mechanism of bubbles nucleated in the PP/PS/CO₂ blend system, Figure 7 depicts the morphologies of critical bubbles nucleated in different domains. It is clear that the bubble nucleated in the PP domain shown in Figure 7a is characterized with a large size and big contact angle, the bubble nucleated in the PS domain displayed in Figure 7b is characterized with a much smaller size and contact angle, and the bubbles nucleated in the blend interfacial region shown in Figure 7c,d are characterized with further shrinkage of size and contact angle. The contact angle of bubbles is the competition result among the three interfacial tensions of melt–interface, gas–melt, and gas–interface, which directly reflects the wettability of CO₂ in different domains. The contact angles of bubbles nucleated in PP-PP/CO₂ and PS-PS/CO₂ domains are 118 and 108°, respectively. It is much larger than 90°, implying a difficult wettability for CO₂ in PP or PS domains. However, the contact angles of bubbles nucleated in PS-PP/CO₂ and PP-PS/CO₂ domains are 85 and 78°, respectively. It is less than 90°, which provides direct evidence for better wettability of CO₂ in PP and PS contact interfacial regions. As shown, bubbles nucleated in the blend interface attached to the PS-rich phase displayed in Figure 7d achieve the smallest size and contact angle. In other words, the interfacial regions attached to the PS-rich domain perform the best wettability for CO₂ molecules, which provide the possibility for the formation of the smallest bubbles. The results are consistent with the interfacial CO₂ enrichment over the blend interface, which highly induces the thermodynamic instability and enhances bubble nucleation.

Figure 8 shows the nucleation free-energy barriers and critical radii for bubbles nucleated in PP or PS domains and the blend interfacial region. It is noticeable that both the energy barrier and the critical radius decrease with increasing pressure, and the decrease rate varies obviously with the pressure in the beginning. This is mainly attributed to the improving amplitude of local supersaturation that declines at high pressure, resulting in the slow decline of energy and critical radius. Comparing the curves in the figure, one can see that the energy barrier and critical radius for bubbles nucleated in the PP domain are larger than those in the PS domain, and the values for bubbles nucleated in the blend interface attached to the PS-rich phase achieve the smallest ones.

Figure 9 displays the nucleation number densities for bubbles nucleated in different domains. One can find that the predicted nuclei number increases with pressure, the nucleation number densities of bubbles in the PP domain can only reach 3.25×10^6 cell/cm³ at a given pressure, the nucleation number density of bubbles in the PS domain is about two orders of magnitude larger than the value in the PP domain at 20 MPa, and the nucleation number density in the PP/PS blend interfacial region can increase five to six orders of magnitude. There is no doubt that the addition of PS in PP effectively improves the foaming performance of the PP system. During the foaming process, the PP domains serve as a CO₂ reservoir, the blend interfacial regions attached to the PS-rich domain provide the best favorable heterogeneous nucleation sites, and the CO₂ molecules accumulate in the blend interface, resulting in a much higher local CO₂ supersaturation, which makes bubble nucleation with obviously

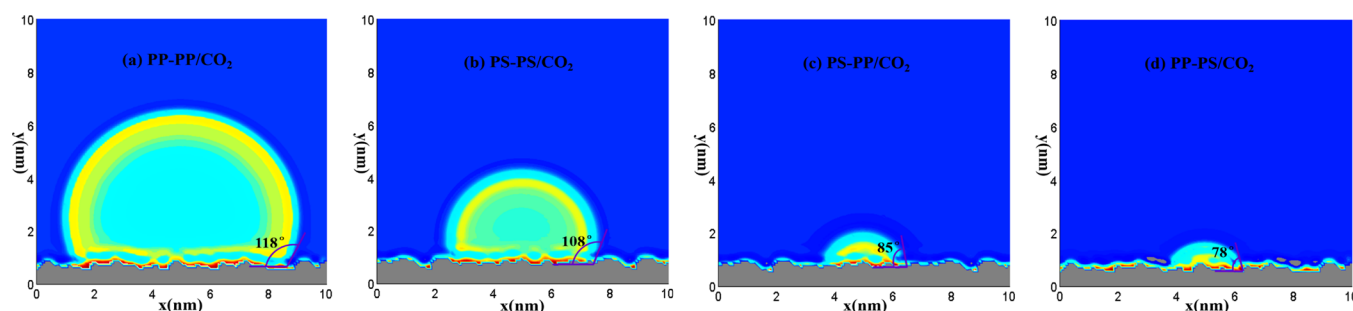


Figure 7. Morphologies of critical CO_2 bubbles nucleated on different surfaces in the PP/PS/ CO_2 blend system, where the profiles cut at $y = 12.8\sigma_{\text{CO}_2}$. (a) PP domain, (b) PS domain, (c) PP and PS blend interface attached to the PP-rich domain, and (d) PP and PS blend interface attached to the PS-rich domain. The mass fractions of CO_2 and polymers are determined at 19.0 MPa.

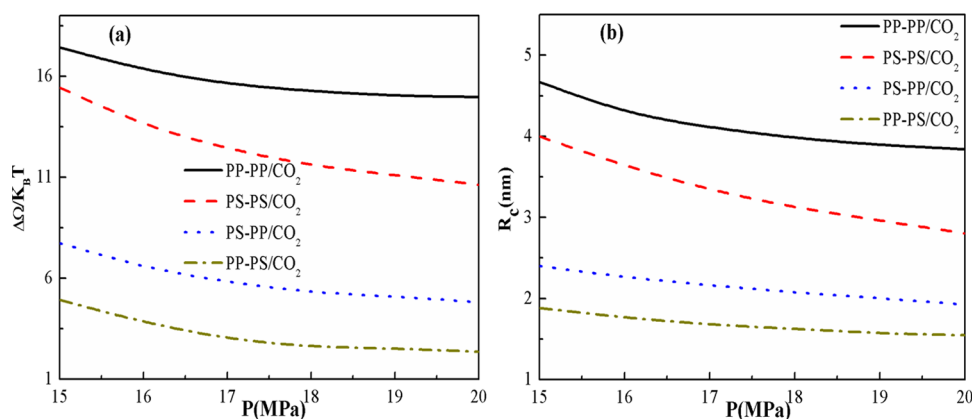


Figure 8. (a) Nucleation free-energy barriers and (b) critical radii of bubbles nucleated in different domains.

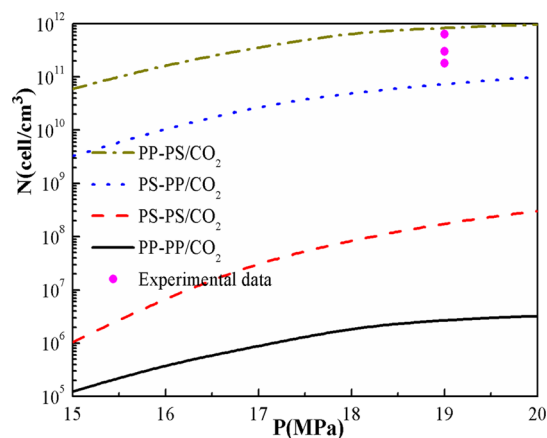


Figure 9. Nucleation number densities of bubbles nucleated in different domains, the curves are predicted values, and the points are experimental data.³⁸

decreasing cell size and dramatically increasing cell density possible. The results are partially examined by the available experimental data and the deviation between theoretical predictions and experimental measurements³⁸ within one order of magnitude. Considering that there exists bubble rupture and coalescence in polymer foaming experiments,⁴¹ we think that the current theoretical model provides a reasonable evaluation of heterogeneous bubble nucleation rates in the PP/PS blend system.

4. CONCLUSIONS

In summary, the foaming mechanism of the PP/PS/ CO_2 blend system is investigated within the constructed 3D-DFT approach. The predicted values reveal that more CO_2 molecules are dissolved in the PP domain than in the PS domain, which will provide a reserve for subsequent bubble nucleation. After decompression, CO_2 molecules tend to migrate from the PP region to the PP/PS blend interface and finally to the PS region, as explained by the interfacial density profiles. The CO_2 enrichment in the blend interfacial region greatly reduces the interfacial tension, which provides more powerful thermodynamic driving force for bubble nucleation. The reduced contact angles, free-energy barriers, critical radii, and nucleation number densities give a direct insight into the nucleation capacity of the bubbles in different regions. Apart from providing a simple, fast, and flexible way to derive various results that are difficult to measure from experiments, the general theoretical framework successfully offer a mechanistic interpretation for blend polymer system foaming.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.1c03087>.

Calculation of excess free energy, construction of the PP and PS surface, external potential of the polymer matrix, and calculation of the pre-exponential factor (PDF)

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Author Contributions

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Notes

The authors declare no competing financial interest.

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